SCSI1113 – COMPUTATIONAL MATHEMATICS

2015/2016 – Semester 2

TEST 2

Instruction: **Answer all questions in the answer booklet.**

QUESTION 1 **[25 MARKS]**

i) Let, **w** be the subset of vector space of 2x2 matrices.

$$w = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \middle| a + b = 5 \right\}$$

Check whether **w** is subspace under the inherited operations. (3 marks)

ii) Let, $B_1 = \{v_1, v_2\}$ and $B_2 = \{w_1, w_2\}$ be bases for a vector space, V = $\mathbf{R^2}$, and suppose

$$v_1 = 2w_1 + 3w_2,$$
$$v_2 = -w_2.$$

a) State the transition matrix from B_1 to B_2, $[I]_{B_1}^{B_2}$. (2 marks)

b) If $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, find w_1 and w_2. (5 marks)

c) Find $[v]_{B_1}$ if $[v]_{B_2} = \begin{pmatrix} 6 \\ 11 \end{pmatrix}$. (3 marks)

Hint: $[v]_{B_2} = [I]_{B_1}^{B_2} [v]_{B_1}$

iii) Matrix ***A*** is given as follows.

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 \end{bmatrix}$$

a) State the row space of ***A***. (2 marks)

b) Find the nullspace of ***A***. (6 marks)

c) Find the basis of column space of ***A.*** (2 marks)

d) Find the rank and nullity of ***A.*** (2 marks)

QUESTION 2 **[10 MARKS]**

i) The function $T : R^3 \to R^2$ is defined as

$$T(v) = A(v) = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

a) Find the image of $v = \begin{bmatrix} -5 \\ 7 \\ -3 \end{bmatrix}$ (2 marks)

b) Find the kernel of the linear transformation, ker(T) (4 marks)

ii) Let $T : R^3 \to R^3$ be a linear transformation such that

$$T(1,0,0) = (-2,-1,4)$$
$$T(0,1,0) = (1,-5,3)$$
$$T(0,0,1) = (6,0,-5)$$

Find $T(7, -1, 4)$ (4 marks)

Hint: $v = c_1 v_1 + c_2 v_2 + c_3 v_3$

$$T(v) = c_1 T(v_1) + c_2 T(v_2) + c_3 T(v_3)$$

QUESTION 3 **[25 MARKS]**

Given the function T:$R^2 \rightarrow R^3$ defined as

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ 2x_2 - x_1 \\ 4x_1 \end{bmatrix}$$

i) Show that T is a linear transformation. (8 marks)

ii) Find the matrix representation $[T]_B^{B'}$ for T relative to the bases below:

$$B_1 = \left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix}\right\} \text{ for } R^2 \text{ and } B_2 = \left\{\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}\right\} \text{ for } R^3$$

(11 marks)

iii) Let $v = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$. Find T(v) use matrix found in 2(ii). (6 marks)